Global Malaria Burden Analysis

An Epidemiological Assessment Using WHO Data

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Tools Used

Python • Pandas • Matplotlib • Plotly

Jupyter Notebook • Power BI

Data Source

World Health Organization (WHO)

Global Health Observatory (GHO)

Independent Portfolio Project

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1. Introduction

Malaria remains one of the world's most significant vector-borne infectious diseases, posing a major public health challenge, particularly in tropical and subtropical regions. Despite decades of global control efforts, malaria continues to cause substantial morbidity and mortality, disproportionately affecting vulnerable populations, especially children under five years of age and pregnant women in low- and middle-income countries.

According to the World Health Organization (WHO), malaria continues to place a considerable burden on health systems, economies, and communities worldwide. Continuous surveillance and timely analysis of malaria data are essential for monitoring disease trends, identifying high-risk populations, evaluating the effectiveness of control interventions, and supporting evidence-based public health decision-making.

This project presents an end-to-end epidemiological analysis of global malaria data obtained from the World Health Organization (WHO). Using Python for data preprocessing, exploratory data analysis (EDA), and visualization, together with Microsoft Power BI for interactive dashboard development, the project explores temporal trends, geographical distribution, and key malaria indicators across countries and WHO regions.

The findings of this analysis provide valuable insights into the global distribution of malaria and demonstrate how data analytics can support disease surveillance, resource allocation, and strategic planning for malaria prevention and control. In addition, this project highlights practical applications of epidemiological methods, data visualization, and business intelligence tools in addressing real-world public health challenges.

2. Project Objectives

a/ General Objective

The primary objective of this project is to analyze the global burden of malaria using epidemiological data obtained from the World Health Organization (WHO). The project aims to apply data analytics techniques to identify temporal trends, geographical patterns, and key malaria indicators, while demonstrating the use of Python and Microsoft Power BI for public health data analysis and visualization.

b/ Specific Objectives

- To explore the structure and quality of the WHO malaria dataset.
- To clean and preprocess the data to ensure accuracy and consistency for analysis.
- To perform exploratory data analysis (EDA) to summarize key epidemiological characteristics.
- To analyze temporal trends in malaria burden across different reporting years.
- To compare malaria burden across countries and WHO regions.
- To identify countries and regions with the highest reported malaria burden.
- To develop an interactive Power BI dashboard for visualizing malaria indicators.
- To generate epidemiological insights that support evidence-based public health decision-making.
- To provide practical recommendations for strengthening malaria surveillance and control based on the analytical findings.

3. Dataset Description

3.1 Data Source

The dataset used in this project was obtained from the World Health Organization (WHO) through the Global Health Observatory (GHO). It contains internationally reported malaria indicators that support monitoring, surveillance, and evidence-based public health decision-making. The WHO compiles these data from national health information systems and validated reporting mechanisms, making them a reliable source for global epidemiological analysis.

3.2 Dataset Overview

Attribute	**Description**
Dataset Name	Global Malaria Dataset
Source	World Health Organization (WHO)
File Format	CSV
Geographical Coverage	Global
Disease	Malaria
Analysis Tools	Python, Jupyter Notebook, Power BI

3.3 Main Variables

The dataset contains several variables describing malaria indicators, geographical information, and reporting characteristics. The most relevant variables used in this analysis are summarized below.

Variable	**Description**
Indicator	Name of the malaria indicator being reported
FactValueNumeric	Numerical value of the indicator
Location	Country or territory
ParentLocation	WHO Region
Period	Reporting year
DataSource	Source of the reported data
ValueType	Type of reported value
FactValueUoM	Unit of measurement

3.4 Data Structure

The dataset consists of observations reported for different countries, WHO regions, years, and malaria indicators. Each row represents one observation for a specific indicator, country, and reporting period, while the columns provide descriptive and quantitative information used for epidemiological analysis.

3.5 Data Quality Considerations

Before analysis, the dataset was assessed for data quality. The following preprocessing steps were performed:

- Assessment of missing values.
- Detection and removal of duplicate records.
- Verification of data types.
- Cleaning of text fields.
- Standardization of selected variable names.
- Validation of the cleaned dataset before analysis.

4. Methodology

4.1 Study Design

This project employed a descriptive epidemiological data analysis using secondary malaria surveillance data obtained from the World Health Organization (WHO). The analysis focused on describing the distribution and trends of malaria indicators across countries, WHO regions, and reporting years. A combination of statistical analysis, data visualization, and business intelligence techniques was applied to transform raw data into meaningful public health insights.

4.2 Analytical Workflow

The project followed a structured workflow consisting of six major stages:

- Data Acquisition
- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development
- Epidemiological Interpretation and Reporting

The workflow is illustrated below.

```
WHO Malaria Dataset
      |
      ▼
Data Loading
      |
      ▼
Data Cleaning & Preprocessing
      |
      ▼
Exploratory Data Analysis (EDA)
      |
      ▼
Statistical Analysis & Visualization
      |
      ▼
Power BI Dashboard Development
      |
```

▼

Epidemiological Interpretation

|

▼

Public Health Recommendations

4.3 Software and Tools

The following software and libraries were used throughout the project.

Tool	Purpose
Python	Data analysis and preprocessing
Jupyter Notebook	Interactive analysis and documentation
Pandas	Data manipulation and cleaning
NumPy	Numerical computations
Matplotlib	Statistical visualizations
Seaborn	Advanced data visualization
Plotly	Interactive visualizations
Power BI	Interactive dashboard development
GitHub	Version control and portfolio presentation

4.4 Data Cleaning Procedures

The dataset was prepared for analysis through several preprocessing steps, including:

- Inspecting the dataset structure and variable types.
- Identifying missing values and assessing their impact.
- Detecting and removing duplicate records where applicable.
- Standardizing selected variable names for consistency.
- Verifying numerical variables used in statistical analyses.
- Saving a cleaned dataset for downstream analysis and dashboard development.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the dataset and identify meaningful epidemiological patterns. The analyses included:

- Dataset dimensions and structure.
- Number of reporting countries.
- Number of WHO regions.
- Time coverage.
- Distribution of malaria indicators.
- Descriptive statistics.
- Comparison of malaria burden across countries and regions.
- Temporal trend analysis.

4.6 Data Visualization

Several visualizations were developed to communicate analytical findings effectively, including:

- ❖ Bar charts
- ❖ Line charts
- ❖ Histograms
- ❖ Box plots
- ❖ Geographic maps
- ❖ Regional comparison charts

These visualizations were used to identify trends, compare geographical areas, and summarize the distribution of malaria indicators.

4.7 Dashboard Development

An interactive dashboard was developed using Microsoft Power BI to enable dynamic exploration of the malaria dataset. The dashboard consists of four analytical pages:

- Executive Dashboard
- Geographic Analysis
- Trends & Forecasting
- Public Health Insights & Recommendations

Interactive filters (slicers) were included to allow users to explore the data by year, country, WHO region, and malaria indicator.

5. Data Cleaning

5.1 Overview

Data cleaning is a critical step in the data analysis process, as it ensures the accuracy, consistency, and reliability of the dataset. Before conducting the epidemiological analysis, the malaria dataset was examined for data quality issues, including missing values, duplicate records, inconsistent data types, and formatting problems. Appropriate preprocessing techniques were applied to prepare the dataset for statistical analysis and visualization.

5.2 Missing Value Assessment

The dataset was inspected to identify missing values across all variables. Missing data were assessed to determine their potential impact on the analysis.

The following Python command was used:

Python Run

```python
df_clean.isnull().sum()
```

The assessment showed that the majority of variables contained complete information. Any missing values were reviewed to ensure they would not compromise the analytical results.

5.3 Duplicate Records

Duplicate observations were identified to prevent double counting during statistical analysis.

The following command was used:

```python
df_clean.duplicated().sum()
```

5.4 Data Type Verification

Variable data types were examined to confirm that numerical variables were stored as numeric values and categorical variables as text.

The following command was used:

```python
df_clean.info()
```

Variables used in statistical calculations, particularly **FactValueNumeric**, were verified to have appropriate numeric data types.

6. Exploratory Data Analysis (EDA)

6.1 Overview

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and characteristics of the malaria dataset before developing visualizations and dashboards. The analysis focused on summarizing the dataset, identifying epidemiological patterns, and answering key public health questions related to malaria burden across countries, WHO regions, and reporting years.

6.2 Dataset Summary

The initial exploration examined the overall structure of the dataset, including the number of observations, variables, and data types.

The following Python commands were used:

```python
df_clean.shape
df_clean.info()
df_clean.describe()
```

These summaries provided an overview of the dataset and confirmed that the variables required for analysis were available and correctly formatted.

6.3 Geographic Coverage

The analysis identified the number of countries and WHO regions represented in the dataset.

Example analyses included:

- Number of reporting countries.
- Number of WHO regions.
- Distribution of records across geographical areas.

This assessment provides an understanding of the geographical scope of the malaria surveillance data.

6.4 Temporal Coverage

The reporting period covered by the dataset was examined to understand the time span available for trend analysis.

Key questions addressed include:

- What is the earliest reporting year?
- What is the most recent reporting year?
- How many years are represented in the dataset?

Understanding the temporal coverage is essential for evaluating changes in malaria burden over time.

6.5 Distribution of Malaria Indicators

The analysis examined the malaria indicators available in the dataset and their frequency of occurrence.

This helped identify:

- Available malaria indicators.
- Frequency of reporting.
- Indicators selected for further analysis.

6.6 Descriptive Statistics

Descriptive statistics were calculated for the numerical variables used in the analysis.

These statistics included:

- ❖ Mean
- ❖ Median
- ❖ Minimum
- ❖ Maximum
- ❖ Standard deviation
- ❖ Quartiles

These measures summarize the overall distribution and variability of malaria burden values.

6.7 Top Countries by Malaria Burden

Countries were ranked according to their reported malaria burden to identify those contributing the largest share of reported cases or other selected indicators.

This analysis highlights geographical disparities and helps identify priority countries for malaria control and surveillance.

6.8 WHO Region Comparison

Malaria burden was compared across WHO regions to identify regional differences in disease distribution.

The comparison supports the identification of regions requiring sustained public health interventions and resource allocation.

6.9 Temporal Trend Analysis

Temporal analysis was performed by aggregating malaria burden across reporting years.

This analysis helps determine whether malaria burden increased, decreased, or remained stable over time and provides insight into the effectiveness of malaria control efforts.

6.10 Key Findings

The exploratory analysis revealed several important epidemiological observations:

The dataset includes malaria information from multiple countries and WHO regions.

Considerable geographical variation exists in malaria burden.

Temporal patterns demonstrate changes in malaria burden across reporting years.

A relatively small number of countries contribute a substantial proportion of the reported malaria burden.

The dataset provides valuable information for surveillance, planning, and evidence-based public health decision-making.

Note: Adjust these findings to reflect the actual results from your analysis rather than making claims not supported by your data.

7. Data Visualization

7.1 Overview

Data visualization plays a crucial role in epidemiological analysis by transforming complex datasets into clear and interpretable graphical representations. In this project, multiple visualization techniques were used to explore the temporal trends, geographical distribution, and comparative burden of malaria across countries and WHO regions.

The visualizations were developed using Matplotlib, Seaborn, and Plotly in Python, and were later integrated into an interactive Power BI dashboard.

7.2 Distribution of Malaria Burden

A histogram was used to examine the distribution of malaria burden values.

- **Purpose**

Assess the distribution of reported malaria values.

Identify skewness and extreme observations.

Understand the overall variability of the dataset.

- **Interpretation**

The histogram provides an overview of how malaria burden values are distributed and helps identify whether the data are normally distributed or concentrated within specific ranges.

7.3 Top 10 Countries by Malaria Burden

A horizontal bar chart was created to identify the countries reporting the highest malaria burden.

- **Purpose**

Compare malaria burden among countries.

Identify priority countries for surveillance and intervention.

Highlight geographical disparities.

❖ **Interpretation**

The chart ranks countries according to the selected malaria indicator, allowing rapid identification of those contributing the largest share of the reported burden.

7.4 Temporal Trend Analysis

A line chart was used to examine changes in malaria burden over time.

❖ **Purpose**

Evaluate temporal trends.

Identify increases or decreases in malaria burden.

Support assessment of long-term epidemiological patterns.

❖ **Interpretation**

The line chart illustrates how the malaria burden changed across reporting years and provides insight into the effectiveness of malaria prevention and control efforts.

7.5 WHO Region Comparison

A bar chart was developed to compare malaria burden among WHO regions.

❖ **Purpose**

Compare disease burden across WHO regions.

Identify regions with the greatest malaria burden.

Support regional prioritization.

- ❖ **Interpretation**

Regional comparisons highlight geographical differences in malaria burden and help identify areas requiring sustained public health interventions.

7.6 Geographic Distribution

A world map was used to visualize the geographical distribution of malaria burden.

- ❖ **Purpose**

Display spatial variation.

Identify high-burden countries.

Improve interpretation through geographical context.

- ❖ **Interpretation**

The map demonstrates the unequal distribution of malaria burden across the world and supports the identification of high-priority countries for surveillance and control.

7.7 Dashboard Visualizations

The visualizations developed in Python were complemented by an interactive Power BI dashboard containing four analytical pages:

- Executive Dashboard
- Geographic Analysis
- Trends & Forecasting
- Public Health Insights & Recommendations

The dashboard enables users to interactively explore the data using filters for:

- Year
- Country
- WHO Region

- Malaria Indicator

This interactive design supports flexible analysis and enhances decision-making by allowing users to focus on specific geographical areas, time periods, or indicators.

8. Dashboard Overview

8.1 Overview

To enhance the interpretation and communication of epidemiological findings, an interactive dashboard was developed using Microsoft Power BI. The dashboard provides a user-friendly interface for exploring the global malaria burden through dynamic visualizations and filters. It enables users to analyze malaria indicators across countries, WHO regions, and reporting years, supporting evidence-based public health decision-making.

The dashboard was designed with the principles of clarity, interactivity, and accessibility, allowing users to quickly identify trends, compare regions, and explore key epidemiological indicators.

8.2 Dashboard Structure

The dashboard consists of four interactive pages, each focusing on a different aspect of the analysis.

Page	**Description**
Executive Dashboard	High-level summary of the global malaria burden using key performance indicators (KPIs) and summary visualizations.
Geographic Analysis	Interactive maps and charts illustrating the geographical distribution of malaria across countries and WHO regions.
Trends & Forecasting	Time-series visualizations showing changes in malaria burden over time and supporting trend assessment.

8.3 Dashboard Design Principles

The dashboard was designed according to the following principles:

- Clarity**:** Simple and intuitive layout for easy interpretation.
- Consistency: Uniform color palette, typography, and formatting across all pages.
- Interactivity: Dynamic filtering to support flexible exploration.
- Accessibility: Clear labels, legends, and titles for all visualizations.
- Decision Support: Focus on presenting actionable epidemiological information.

9.1 Overview

The epidemiological analysis of the WHO malaria dataset provided valuable insights into the global distribution and temporal patterns of malaria. By combining descriptive statistics, data visualization, and interactive dashboard exploration, several important findings were identified regarding the burden of malaria across countries and WHO regions.

9.2 Geographic Distribution

The analysis demonstrated that the burden of malaria is not evenly distributed across the world. Instead, a relatively small number of countries account for a substantial proportion of the reported malaria burden.

This geographical concentration highlights the importance of prioritizing high-burden countries for surveillance, prevention, diagnosis, treatment, and vector control interventions.

9.3 Regional Differences

Comparison across WHO regions revealed considerable variation in malaria burden. These differences may reflect variations in:

- Climate and environmental conditions.
- Mosquito vector distribution.
- Access to healthcare services.
- Coverage of malaria prevention and control programmes.
- Strength of disease surveillance systems.

Understanding these regional differences is essential for planning targeted public health interventions.

9.4 Temporal Trends

The time-series analysis showed that malaria burden changed across the reporting years included in the dataset.

Temporal analysis is important for:

- Monitoring progress toward malaria control goals.
- Detecting periods of increased transmission.
- Evaluating the effectiveness of public health interventions.
- Supporting long-term surveillance

Replace this paragraph with your actual findings, for example:

"The reported malaria burden declined over the study period."

"The burden increased during specific years before stabilizing."

9.5 High-Burden Countries

Ranking countries by malaria burden identified those contributing the greatest proportion of reported cases (or the selected indicator).

These countries should remain priorities for:

- Enhanced disease surveillance.
- Vector control programmes.
- Early diagnosis and effective treatment.
- Resource allocation.
- Monitoring and evaluation.

9.6 Public Health Implications

The findings demonstrate the value of epidemiological surveillance data for guiding evidence-based public health decisions. Reliable surveillance systems allow health authorities to:

- Monitor disease trends.
- Identify populations at greatest risk.
- Allocate healthcare resources efficiently.
- Evaluate malaria control programmes.
- Support national and international malaria elimination strategies.

9.7 Key Findings

The analysis identified the following major findings:

- The malaria burden is unevenly distributed across countries and WHO regions.
- Geographic disparities indicate the need for targeted malaria control strategies.
- Temporal analysis provides evidence of changes in malaria burden over the reporting period.
- A relatively small number of countries contribute a large share of the reported burden.
- Interactive dashboards improve the communication of epidemiological evidence and support data-driven decision-making.

9.8 Discussion

The results of this analysis are consistent with the importance of continuous malaria surveillance in supporting public health planning. Although global malaria control efforts have contributed to reducing disease burden in many settings, substantial challenges remain in high-transmission areas. Data-driven approaches, including interactive dashboards and epidemiological analysis, can improve the identification of priority populations and strengthen monitoring of malaria control programmes.

10. Public Health Recommendations

10.1 Overview

The findings of this epidemiological analysis provide valuable evidence for strengthening malaria prevention, surveillance, and control efforts. Based on the observed temporal and geographical patterns, the following recommendations are proposed to support public health decision-making and improve malaria control strategies.

10.2 Strengthen Malaria Surveillance Systems

Continuous, high-quality surveillance is essential for monitoring malaria trends and detecting changes in disease transmission. National malaria programmes should strengthen routine surveillance systems by improving the completeness, accuracy, and timeliness of case reporting.

Priority actions include:

- Improving electronic disease reporting systems.
- Strengthening routine data quality assessments.
- Expanding surveillance coverage in underserved areas.
- Enhancing data validation procedures.

10.3 Prioritize High-Burden Countries and Regions

The analysis identified substantial geographical variation in malaria burden, indicating that interventions should be prioritized in areas with the highest reported burden.

Recommended actions include:

- Targeting resources to high-burden settings.
- Increasing coverage of malaria prevention programmes.
- Strengthening community-based surveillance.
- Improving access to diagnosis and treatment services.

10.4 Improve Data Quality

Reliable epidemiological analyses depend on high-quality surveillance data.

Health authorities should:

- Standardize data collection procedures.
- Reduce missing and inconsistent records.
- Conduct regular data quality audits.
- Train healthcare workers in surveillance reporting.

Improved data quality will increase the reliability of future epidemiological analyses and public health decision-making.

10.5 Expand Preventive Interventions

Malaria prevention remains one of the most effective strategies for reducing disease burden.

Public health programmes should continue to strengthen:

- Insecticide-treated net (ITN) distribution.
- Indoor residual spraying (IRS).
- Early diagnosis and prompt treatment.
- Community health education.
- Integrated vector management programmes.

11. Limitations

11.1 Overview

While this project provides valuable insights into the global burden of malaria, several limitations should be considered when interpreting the findings. These limitations are primarily related to the nature of the data source, the scope of the analysis, and the available variables.

11.2 Secondary Data Source

The analysis was based entirely on secondary data obtained from the World Health Organization (WHO). Although WHO data are internationally recognized and widely used, the quality of the dataset depends on the completeness and accuracy of reporting by individual countries.

11.3 Data Quality and Reporting

The reported malaria indicators may be affected by:

- Differences in national surveillance systems.
- Variations in reporting practices.
- Delays in data submission.
- Changes in case definitions or reporting methodologies over time.

These factors may influence comparisons between countries or across different reporting years.

11.4 Limited Variables

The dataset primarily contains surveillance indicators and does not include several factors known to influence malaria transmission, such as:

- Climatic conditions (e.g., rainfall and temperature).
- Population density.
- Socioeconomic status.
- Healthcare accessibility.
- Intervention coverage (e.g., insecticide-treated nets and indoor residual spraying).
- Mosquito vector distribution.

Including these variables could provide a more comprehensive understanding of malaria epidemiology.

11.5 Descriptive Nature of the Analysis

This project focused on descriptive epidemiological analysis and data visualization. It was designed to summarize patterns and trends rather than establish causal relationships between malaria burden and potential risk factors.

11.6 Generalizability

The findings presented in this report are specific to the available WHO malaria dataset and the selected indicators included in the analysis. Results should therefore be interpreted within the context of the dataset and should not be generalized beyond the available data without additional supporting evidence.

11.7 Future Improvements

Several enhancements could strengthen future versions of this project, including:

- Integration of climate and environmental datasets.
- Inclusion of demographic and socioeconomic indicators.
- Analysis at subnational (district or provincial) levels where data are available.
- Development of predictive models using machine learning techniques.

12. Conclusion

12.1 Conclusion

This project presented an end-to-end epidemiological analysis of the global malaria burden using surveillance data obtained from the World Health Organization (WHO). By applying Python for data preprocessing, exploratory data analysis, and statistical visualization, and Microsoft Power BI for interactive dashboard development, the project demonstrated a comprehensive workflow for analyzing and communicating public health data.

The analysis provided valuable insights into the geographical distribution and temporal trends of malaria, highlighting variations across countries and WHO regions. The interactive dashboard further enhanced data exploration by enabling users to examine malaria indicators dynamically through filtering and visualization, thereby supporting evidence-based public health decision-making.

Beyond the analytical results, this project demonstrates the practical integration of epidemiological principles with modern data analytics tools. It illustrates how routinely collected surveillance data can be transformed into meaningful information that supports disease monitoring, resource allocation, and strategic planning for malaria prevention and control.

Overall, this project reflects competencies in epidemiological data analysis, data management, visualization, and dashboard development. It also highlights the value of combining programming and business intelligence tools to address real-world public health challenges. The methodology presented in this report can be adapted to other infectious diseases and surveillance datasets, making it a valuable framework for future epidemiological and health data analytics projects.

12.2 Project Outcomes

The project successfully achieved the following outcomes:

Conducted a comprehensive epidemiological analysis of the WHO malaria dataset.

Performed data cleaning and preprocessing to improve data quality.

Applied exploratory data analysis to identify temporal and geographical patterns.

Developed informative statistical and epidemiological visualizations.

Designed an interactive four-page Power BI dashboard.

Generated evidence-based public health insights and recommendations.

Produced a reproducible analytical workflow suitable for portfolio presentation and future public health projects.

12.3 Final Remarks

This project demonstrates the importance of integrating epidemiological knowledge with data science techniques to support evidence-based decision-making in global health. As public health surveillance systems continue to generate large volumes of data, professionals with expertise in epidemiology, programming, and data visualization will play an increasingly important role in transforming data into actionable insights. The skills and methods demonstrated in this project provide a strong foundation for future work in epidemiology, disease surveillance, health data analytics, and public health research.

13. References

Include all sources used throughout the project, such as:

1. World Health Organization. *World Malaria Report*. Geneva: WHO.
2. World Health Organization. *Global Health Observatory (GHO) Data Repository*.
3. Microsoft. *Power BI Documentation*.
4. Pandas Development Team. *Pandas Documentation*.
5. NumPy Developers. *NumPy Documentation*.
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7. Seaborn Documentation.
8. Plotly Documentation.
9. Jupyter Project Documentation.

10. Python Software Foundation. *Python Documentation*.